🎼 **Go it is.** Here are the two missing “lock-the-engine” pieces:

- **DriftEligibility()** — when SLOW traits are allowed to move
- **ShadowActivation rules** — deterministic scoring per shadow type, per event

1) DriftEligibility() — SLOW drift gate (v1)

Purpose

Prevent identity chaos. SLOW traits only drift when there’s **stable evidence** that a user has actually changed (or onboarding was inaccurate).

Inputs

- rollup_n50 (last 50 events)
- rollup_n30 (last 30 events)
- current genome
- config thresholds

Outputs

- eligible: bool
- reason_codes: string[]
- confidence: float

Required metrics (from rollups)

Compute:

- E50 escalation_rate over last 50
- E20 escalation_rate over last 20
- R50 repair_success_rate over last 50 (repair_event / tension_events)
- L50 mean repair latency normalized over last 50 repair events (0–1)
- A_slope50 arousal slope over last 50 (linear fit)
- A_std50 arousal volatility over last 50
- C50 mean contradiction severity over last 50
- move_entropy50 (how stable the behavior distribution is)

Move entropy (stability check): Let p_i be

normalized frequency of move categories in last 50 (withdraw/press/escalate/appease/repair/joke).
$H = -\Sigma p_i * \ln(p_i)$
Normalize: $H_{norm} = H / \ln(K)$ where K = number of move categories present.
Lower entropy = more consistent pattern.

Gate logic (must pass ALL)

G0. Evidence minimum

- total_events >= 50
- unique_days_covered >= 10 (to avoid one bad day driving drift)

G1. Not currently unstable

- E20 <= 0.35 *(if too unstable, don't "redefine identity" mid-storm)*
- A_std50 <= 0.22 *(configurable; high volatility = postpone)*

G2. Trend is stable (not just noise)

- abs(A_slope50) <= 0.05 *(if arousal trend is still rising/falling hard, postpone)*

G3. Contradictions aren't spiking

- C50 <= 0.55 OR C50 decreased by >= 0.10 vs previous 50-window

G4. Consistency of behavior

- H_norm <= 0.78 *(behavior distribution not wildly chaotic)*

If all pass → eligible.

Confidence score

Compute a simple eligibility confidence (0–1):

conf = 0 conf += 0.25 * (1 - clamp(E20 / 0.35, 0, 1)) conf += 0.20 * (1 - clamp(A_std50 / 0.22, 0, 1)) conf += 0.20 * (1 - clamp(abs(A_slope50) / 0.05, 0, 1)) conf += 0.20 * (1 - clamp(C50 / 0.55, 0, 1)) conf += 0.15 * (1 - clamp(H_norm / 0.78, 0, 1))

Eligible requires:

- eligible == true AND conf >= 0.60

Pseudocode

function DriftEligibility(roll50, roll20, rollPrev50): reasons = [] if roll50.total_events < 50: reasons += ["insufficient_events"] if roll50.unique_days < 10: reasons += ["insufficient_day_spread"] if roll20.escalation_rate > 0.35: reasons += ["too_unstable_recently"] if roll50.arousal_std > 0.22: reasons += ["high_volatility"] if abs(roll50.arousal_slope) > 0.05: reasons += ["trend_not_settled"] contradiction_ok = (roll50.contradiction_mean <= 0.55) OR ((rollPrev50 exists) AND (rollPrev50.contradiction_mean - roll50.contradiction_mean >= 0.10)) if not contradiction_ok: reasons += ["contradictions_spiking"] if roll50.move_entropy_norm > 0.78: reasons += ["behavior_inconsistent"] eligible = (reasons is empty) conf =

compute_conf(...) return {eligible: eligible AND conf>=0.60, reason_codes: reasons, confidence: conf}

Default v1 recommendation: SLOW drift OFF except shadow_trait_prior micro-drift and attachment_stability_index. Archetype drift stays OFF until you're confident.

2) ShadowActivation Scoring Rules (deterministic v1)

Core idea

Each shadow has:

- a **signature** (trigger + move + state pattern)
- an activation score a_k(t) in **0–1**
- updated into shadow_activation_recent with β-EWMA

You already have:

- withdraw_t, press_t, appease_t (0/1)
- trigger flags: T_rejection, T_ambiguity, T_disrespect, T_control_loss,

T_unfairness, T_shame_exposure
- optional state: arousal_pre, safety_pre, connection_pre
- optional needs: need_reass, need_autonomy, need_clarity
- escalation_event, rupture_event

Helper features

Define:

- threat_t = clip(0.6*(1 - safety_pre) + 0.4*arousal_pre, 0, 1)
- connection_high = 1 if connection_pre >= 0.65 else 0
- safety_low = 1 if safety_pre <= 0.35 else 0
- arousal_high = 1 if arousal_pre >= 0.75 else 0
- reass_high = 1 if (needs_pre.reassurance >= 0.70) else 0 *(if needs exist)*
- clarity_high = 1 if (needs_pre.clarity >= 0.70) else 0

If needs don’t exist, approximate:

- reass_high ≈ press_t AND (T_rejection OR T_ambiguity)
- clarity_high ≈ press_t AND T_ambiguity

Shadow 1: 🧠 Manipulator

Signature: control + outcome steering + deflection when challenged (without open rupture).

Activation score

a = clip(0.35*T_disrespect + 0.25*T_unfairness + 0.20*(press_t) + 0.20*(safety_low AND NOT escalation_event), 0, 1)

Add-on (if you log “evasion markers” like topic shift/word salad):

+0.15*evasion_marker

Note: We keep this conservative to avoid false labeling. It should activate only with repeated patterns.

Shadow 2: 🧱 Stonewaller

Signature: withdrawal + shutdown + control-loss sensitivity.
a = clip(0.45*withdraw_t + 0.25*T_control_loss + 0.15*(arousal_high AND withdraw_t) + 0.15*(T_disrespect AND withdraw_t), 0, 1)

Shadow 3: 💥 Love Bomber

Signature: high pursuit + high reassurance hunger + connection intensity under low safety.
a = clip(0.35*press_t + 0.25*reass_high + 0.20*(connection_high AND safety_low) + 0.20*(T_rejection OR T_ambiguity), 0, 1)

Shadow 4: 🎭 Chameleon

Signature: appeasement + self-erasure under threat + rapid shifting to maintain connection.
a = clip(0.45*appease_t + 0.25*T_shame_exposure + 0.15*(threat_t) + 0.15*(appease_t AND (T_rejection OR

T_disrespect)), 0, 1)

Shadow 5: 🎯 Scorekeeper

Signature: unfairness focus + pressing for “accounting” + re-litigating.

a = clip(0.45*T_unfairness + 0.25*press_t + 0.15*(clarity_high) + 0.15*(T_disrespect), 0, 1)

Optional add-on if you track “history referencing markers” (e.g., “last time”, “always”, “you never”):

+0.10*history_reference_marker

Shadow 6: 📉 Self-Saboteur

Signature: rupture/self-punish loops + shame + pre-emptive collapse.

a = clip(0.35*rupture_event + 0.25*T_shame_exposure + 0.20*(withdraw_t AND T_rejection) + 0.20*(arousal_high AND safety_low), 0, 1)

ShadowActivation update

(fast EWMA)

Use β = 0.06 default (fast but not twitchy):
shadow_activation_recent_k ← shadow_activation_recent_k + β*(a_k - shadow_activation_recent_k)
Daily cap (recommended):

- activation recent per shadow max ±0.12/day

Effective shadow used by RS and forecasting: shadow_effective_k = clip(0.7*shadow_trait_prior_k + 0.3*shadow_activation_recent_k, 0, 1)

Anti-mislabel guard (important)

Even though it's "internal," protect against false positives:
Activation requires persistence before it can influence readiness/compatibility meaningfully:

- If a_k >= 0.70 for **3 of last 10 events,**

then treat as “active”
- Otherwise treat as “present but not dominant”

This prevents one bad fight from branding someone.

Final wiring notes

- These shadow activations feed the **Forecast Engine** heavily.
- They should only modestly influence **Compatibility** until you have outcomes to calibrate coefficients.

If you want the next layer (this is the spicy one): I can write the **Event Labeling Heuristics v1** (how to infer trigger_labels and user_move from raw text when you don’t have explicit tagging), so your system can operate even before full human annotation.